

SEQUENCES, SERIES AND DIFFERENTIAL EQUATIONS

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What we have learned in the previous lecture

- Sequences $\rightarrow$ set of (complex or real) numbers obtained by assigning to each positive integer n a number z_n
- A series is the operation of adding the z_i terms one after the other.
- How can we verify whether a series converges?
 - Comparison test
 - Ratio test
 - Root test
 - Integral test

Today's agenda



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- Power series
 - Convergence
 - Radius of convergence
 - Functions given by power series
- Taylor and McLaurin series
- Exercises!

Power series

A **power series in powers of $z - z_0$** is a series of the form

$$(1) \quad \sum_{n=0}^{\infty} a_n(z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

where z is a complex variable, $a_0, a_1, \dots$ are complex (or real) constants, called the **coefficients** of the series, and z_0 is a complex (or real) constant, called the **center** of the series. This generalizes real power series of calculus.

If $z_0 = 0$, we obtain as a particular case a *power series in powers of z* :

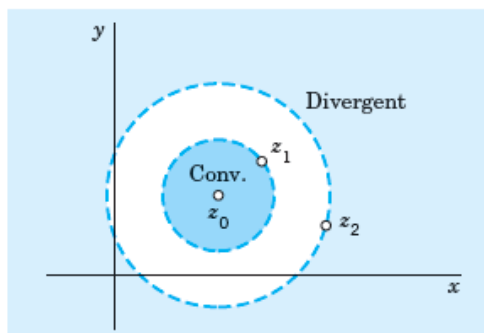
$$(2) \quad \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots$$

Power series - Convergence

- If we fix z , all the concepts for series with constant terms apply.

Convergence of a Power Series

- (a) Every power series (1) converges at the center z_0 .
- (b) If (1) converges at a point $z = z_1 \neq z_0$, it converges absolutely for every z closer to z_0 than z_1 , that is, $|z - z_0| < |z_1 - z_0|$. See Fig. 365.
- (c) If (1) diverges at $z = z_2$, it diverges for every z farther away from z_0 than z_2 . See Fig. 365.

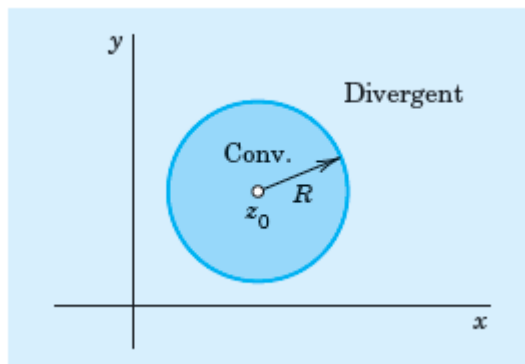


Power series – Radius of convergence

- We consider the **smallest** circle with center that includes all the points at which a given power series converges. Let R denote its radius. The circle

$$|z - z_0| = R$$

is the circle of convergence and R is the radius of convergence.



Power series – Radius of convergence

Radius of Convergence R

Suppose that the sequence $|a_{n+1}/a_n|$, $n = 1, 2, \dots$, converges with limit L . If $L = 0$, then $R = \infty$; that is, the power series (1) converges for all z . If $L \neq 0$ (hence $L > 0$), then

$$(6) \quad R = \frac{1}{L} = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \quad (\text{Cauchy-Hadamard formula}^1).$$

If $|a_{n+1}/a_n| \rightarrow \infty$, then $R = 0$ (convergence only at the center z_0).

Functions given by power series

$$f(z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \dots \quad (|z| < R).$$

- $f(z)$ is represented by the power series.
- A function $f(z)$ cannot be represented by two different power series with the same center $\rightarrow$ uniqueness of a power series representation
- A derived series of a power series is obtained by termwise differentiation.

$$\sum_{n=1}^{\infty} n a_n z^{n-1} = a_1 + 2a_2 z + 3a_3 z^2 + \dots$$

- The derived series of a power series has the same radius of convergence as the original series.
- The integrated series of a power series has the same radius of convergence as the original series.

Taylor and Maclaurin series

- A Taylor series of a function $f(z)$ is

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{where} \quad a_n = \frac{1}{n!} f^{(n)}(z_0)$$

- A Maclaurin series is a Taylor series with center $z_0=0$.

$$\begin{aligned} f(z) = & f(z_0) + \frac{z - z_0}{1} f'(z_0) + \frac{(z - z_0)^2}{2!} f''(z_0) + \cdots \\ & + \frac{(z - z_0)^n}{n!} f^{(n)}(z_0) + R_n(z). \end{aligned}$$

remainder of the Taylor series

Taylor and Maclaurin series

- Geometric series

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \dots$$

- Exponential function

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2} + \dots$$

- Trigonometric function

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = 1 - \frac{z^2}{2} + \frac{z^4}{4} - + \dots$$

- Logarithm

$$\operatorname{Ln}(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - + \dots$$